C++ Introduction (Detailed Guide for PDF)

What is C++?

C++ is a high-level, general-purpose programming language that is an extension of the C language. It supports both procedural and object-oriented programming.

History of C++

- Developed by Bjarne Stroustrup
- Created in 1985
- Extension of C language

Why Use C++?

- High performance
- Used in game development
- System programming
- Competitive programming

Features of C++

- Object-Oriented Programming (OOP)
- Fast execution
- Low-level memory manipulation
- Portable

Basic Program Structure

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!";
    return 0;
}
```

Variables

```
int x = 10;
float y = 5.5;
char ch = 'A';
```

Data Types

- int
- float
- double
- char
- bool

Operators

- Arithmetic (+, -, *, /)
- Relational (==, !=, >, <)
- Logical (&&, ||)

Input and Output

```
int x;
cin >> x;
cout << x;
```

Conditional Statements

```
if (x > 10) {
    cout << "Greater";
} else {
    cout << "Smaller";
}
```

Loops

For Loop

```
for (int i = 0; i < 5; i++) {
    cout << i;
}
```

While Loop

```
int i = 0;
while (i < 5) {
    i++;
}
```

Functions

```
int add(int a, int b) {
    return a + b;
}
```

Arrays

```
int arr[3] = {1, 2, 3};
```

Pointers

```
int x = 10;
int *ptr = &x;
```

Object-Oriented Programming

Class and Object

```
class Person {
public:
    string name;
};
```

Inheritance

Polymorphism

Encapsulation

Constructors

```
class Test {
public:
    Test() {
        cout << "Constructor called";
    }
};
```

File Handling

```
#include <fstream>

ofstream file("test.txt");
file << "Hello";
file.close();
```

Example Program

```
#include <iostream>
using namespace std;

int main() {
    int a = 5, b = 10;
    cout << a + b;
    return 0;
}
```

Advantages of C++

- Fast and efficient
- Supports OOP
- Widely used

Disadvantages of C++

- Complex for beginners
- Manual memory management

Best Practices

- Use meaningful variable names
- Follow proper indentation
- Use comments